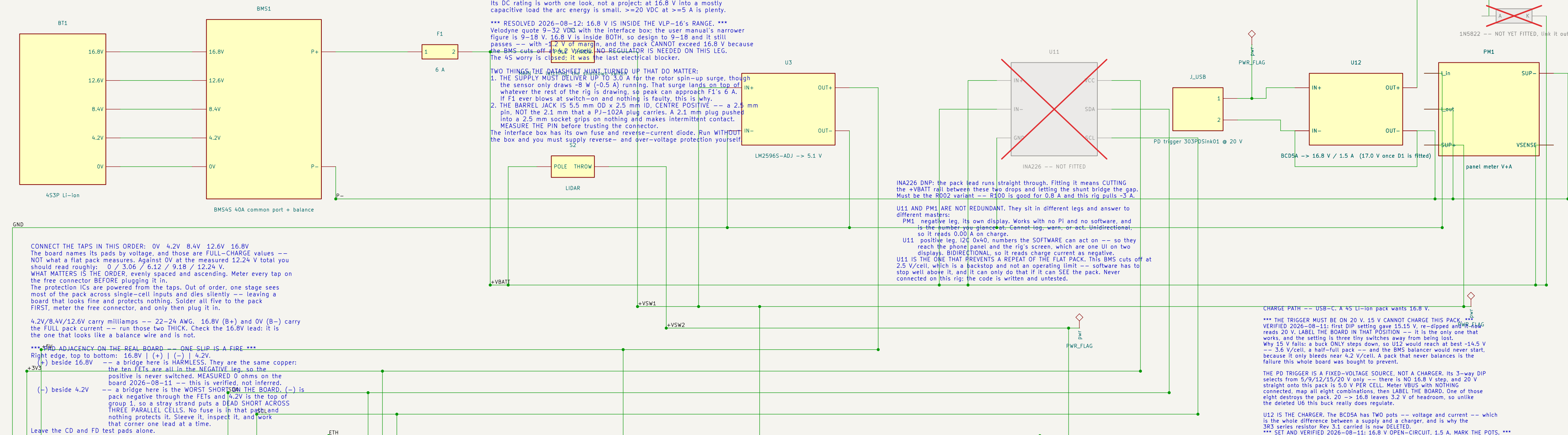
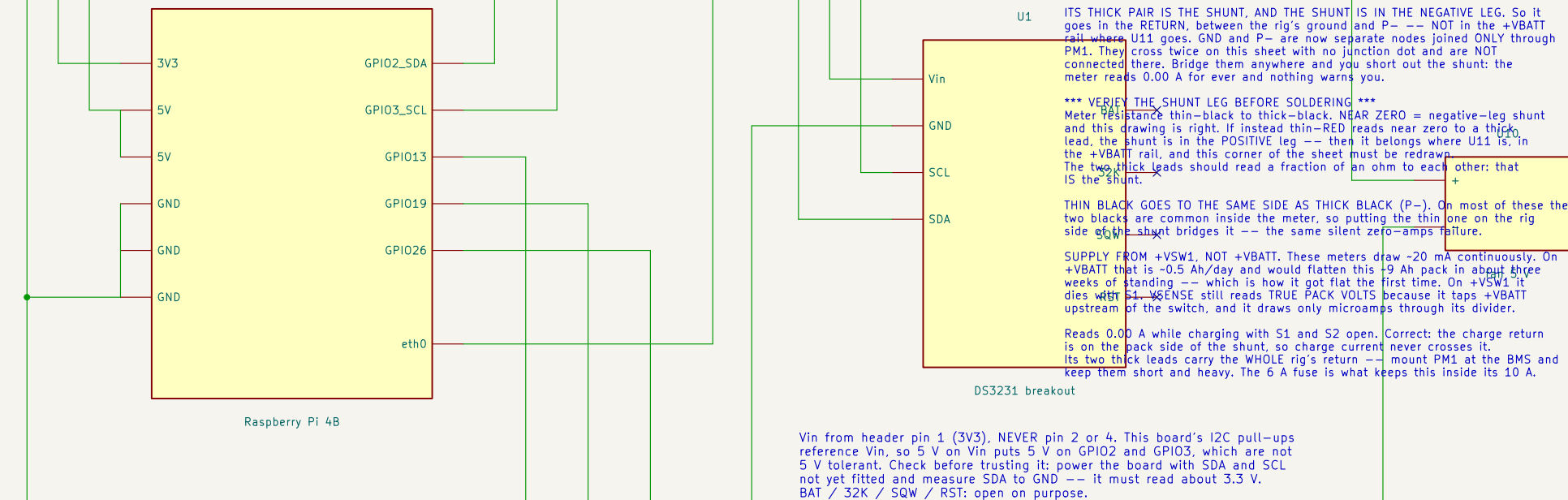


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge



ZONE²P1 THE PI AND ITS BUS



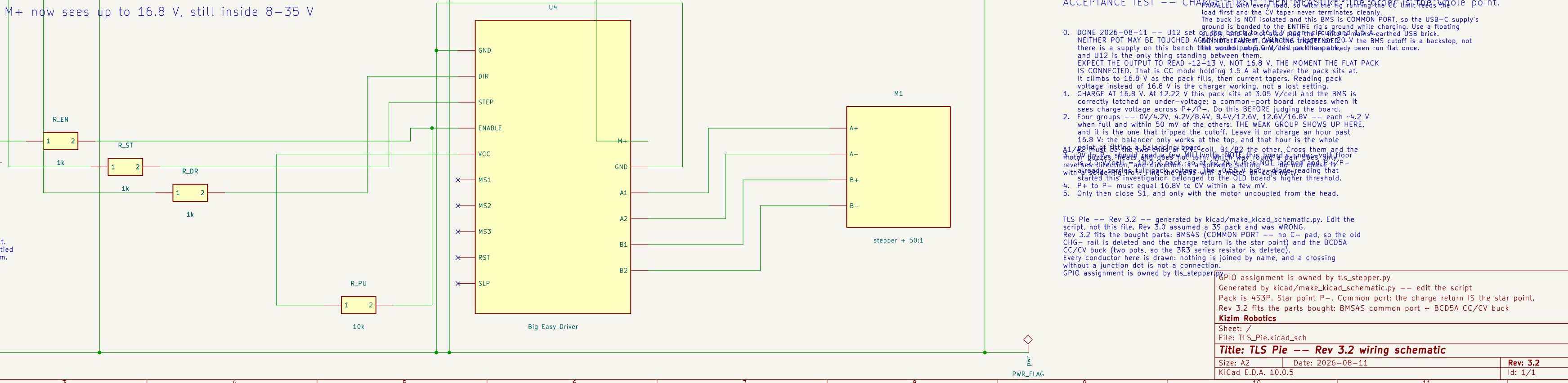
ZONE C MOTOR CHAIN

NO, THE MOTOR DOES NOT NEED A 12 V BUCK. U6 STAYS DELETED.
A stepper's "12 V" rating is just $I_{rated} \times R_{phase}$ --- the volts you would need with NO chopping. U4 is a current-CHOPPING driver: it PWMs the supply to hold the coil at whatever CUR ADJ is set to, so the supply sets how FAST current rises, not how MUCH. More volts is more torque at speed, and the current limit is what protects the motor. 16.8 V is inside U4's 8-35 V.

*** BUT THE MOTOR HAS ONLY EVER RUN ON A FLAT PACK. ***
At 12.2 V the driver may never have reached its setpoint at speed. At 16.8 V it will. Expect MORE TORQUE AND A HOTTER MOTOR on the first charged run even though nothing was adjusted. CHECK THE MOTOR TEMPERATURE, and run it uncoupled from the head first. Do NOT touch CUR ADJ PWR to compensate.

Re-adding U6 would now half-work, which is worse than not working: at 4S it regulates above -13.5 V and drops out below, so the rig would behave one way on a full pack and another on a low one. It would also throw away the torque, add heat and a failure point, and put a -2-3 A module ceiling in front of a chain that draws peaks. The 16.8 V worry belongs to the VLP-16 on S2, not here

R_PU is the gating item — nothing turns until it is fitted. Every PI GPIO floats as an input for the -30 s the PI takes to boot, and ENABLE is active-low. Without this the driver can sit energised through the whole of boot with nothing in control of it. It goes on the DRIVER side of R_EN and pulls up to the driver's own VCC — not the Pi's 3V3 — so it still holds with the PI unplugged. Set the APWR jumper to 3.3 V and MEASURE VCC first. MS1-3 open = 1/16 step, via the driver's own pull-ups. RST and SLP are tied together on-board on a genuine Big Easy Driver — meter them and confirm.



ACCEPTANCE TEST -- CHARGE TEST -- CHARGE WITH S1 AND S2 OPEN. The charger hangs on the fused node in PARALLEL with every load, so with the Hg running the EC limit feeds the whole point.

0. NONE 2026-08-11 -- U12 set output bank to 0V and the battery was recharged by the earthed USB brick.
NEITHER POT MAY BE TOUCHED AGAIN UNTIL THE CHARGING IS COMPLETED - V BMS cutoff is a backup stop, therefore a supply
of 0V will not start charging again until the battery has already been run flat once.
1. EXPECT THE OUTPUT TO READ -12 ~13 V, NOT 16.8 V, AT THE MOMENT THE FLAT PAKK IS CONNECTED. That is CC mode holding at 1.5 A at whatever the pack sits at.
It climbs to 16.8 V as the pack fills, then current takes over and it charges voltage instead of 16.8 V is the charger working, not a lost setting.
2. CHARGE AT 16.8 V. At 12.22 V this packs sits at 3.05 V/cell and the BMS is correctly taking charge; a common-port or relay can be used if needed so it sees charge voltage at +P/-0. DO NOT DEFUDGE JIGGING the board.
3. Four groups -- 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V -- each 4~4.2 V when full and within 50 mV of the cutoff. LEAG GROUP SHOWS UP HERE, and it is the one that tripped the cutoff. Leave it on cutoff for an hour at 16.8 V; the balancer only works at the top, and that hour is the whole
4. point of being a balance group. Cell B4/B2 the other. Cross them and the motor buzzes. Hears and does not turn. (U12) was not a pair before, but now it is. The motor is the best test. If it runs, the battery is good. If it doesn't, the battery is bad. I have seen a lot of people who have tried to fix a battery by putting it back on a charger, but they haven't started this investigation belonged to the OLD board's higher threshold.
5. +P to 0V may equal 16.8V to 0V within a few mV.
6. +P to 0V close S1, and only with the motor uncoupled from the head.

```

TLS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the
script, not this file. Rev 3.0 assumed a 35 pad and was WRONG.
Rev 3.2 fits the bought parts: BMS45 (COMMON PORT -- no C-- pad, so the o
CHG-- rail is deleted and the charge return is the star point) and the BCD5A
CC/CV buck (two pots, so the 3R3 series resistor is deleted).
Every conductor here is drawn: nothing is joined by name, and a crossing
without a junction dot is not a connection.
GPIO assignment is owned by tls_stepper.py

```

GPIO assignment is owned by `tls_stepper.py`
 GPIO assignment is owned by `tls_stepper.py`
 Generated by `kicad/make_kicad_schematic.py` -- edit the script
 Pack is 453P. Star point P-. Common port: the charge return IS the star point.
 Rev 3.2 fits the parts bought: BMS45 common port + BCD5A CC/CV buck

Kizim Robotics
 Sheet: /
 File: `TLS_Pie_Kicad_sch`

Title: TLS Pie -- Rev 3.2 wiring schematic

Size: A2	Date: 2026-08-11	Rev: 3.2
KICad E.D.A. 10.0.5		Id: 1/1